# Mohsin Khawaja

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## Education

**University of California, San Diego** **Aug 2021 – June 2025**

***B.S. Cognitive Science: Machine Learning & Neural Computation, Minor in Computer Science***

**Coursework:** Machine Learning, LLMs, Reinforcement Learning, Computer Vision, Data Structures & Algorithms

**Leadership:** Founder & VP (Muslim Tech Collaborative), Board Member (SDx), Captain (Men's Club Soccer)

## Experience

**Idler (YC S25)** **Mar 2026 – Present**

*Problem Design Engineer* *San Francisco, CA*

- Review production repositories and commit histories to trace engineering intent and refine real-world coding problems.
- Evaluate problem statements, unit tests, and grading criteria for clarity, correctness, and fairness.
- Reproduce agent failures in sandboxes to isolate implementation, test, dependency, and environment issues.

**Aviz Networks** **Oct 2025 – July 2026**

*AI Engineer (Customer Success)* *San Jose, CA*

- Built an autonomous NetOps agent resolving **100+ Zendesk tickets** through MCP orchestration, telemetry parsing, CLI tool-calling, and network troubleshooting workflows.
- Developed **10+ LLM agents** for triage, root cause analysis, config validation, and remediation in SONiC networks.
- Integrated RAG, telemetry, logs, and NCP APIs with structured outputs, guardrails, and prompt routing.
- Deployed production agents with CS and sales teams to support customer workflows and live pre-sales demonstrations.

**Amazon** **June 2025 – Aug 2025**

*Data Analytics (Contract)* *Remote*

- Built an analytics pipeline processing **12K+ employee feedback entries** across global regions.
- Applied NLP sentiment classification (**91% accuracy**), SQL, and persona clustering to identify engagement trends.
- Presented findings to senior HR leadership to inform Q3 support strategy and policy decisions.

**UC Berkeley College of Engineering** **June 2024 – June 2025**

*Machine Learning Engineer* *Berkeley, CA*

- Developed real-time BCI control for robotic exoskeletons with **87% accuracy** and **120 ms latency**.
- Optimized the ML pipeline using PCA and cross-validation, achieving **91% F1** and reducing model size by **40%**.
- Built EEG preprocessing tools with band-pass filtering, ICA, wavelet denoising, and spectral feature extraction.

**Matroid** **June 2023 – Aug 2023**

*Software Engineer Intern* *Palo Alto, CA*

- Developed backend services, APIs, and inference pipelines for computer vision in industrial automation and monitoring.
- Shipped production features improving reliability of vision-based detection and scalable image processing.

## Projects

**AutoApply – Autonomous Job Application Agent** | *Python, Playwright, SQLite, Claude API*

- Indexed **26K+ jobs** across **4 platforms**; processed **3.2K applications** and auto-submitted **50+** using browser automation per session.
- Designed tiered form resolution using profile mapping, cached answers, and LLM fallback with local-model failover.
- Engineered browser recovery, application timeouts, and answer-validation guards; covered workflows with **195 tests**.

**EvoScope – AI Trading System** | *PyTorch, RL*

- Developed an LSTM + DQN trading system achieving **94.2% accuracy** and a **2.52 Sharpe ratio**.
- Built trade execution across Binance, Coinbase, and EVM-compatible DEXs with volatility-based sizing and risk controls.

**LLM Political Evaluation** | *NLP, Bias Detection*

- Designed a political bias evaluation framework across **5+ LLMs** using **100+ controlled prompts**.
- Measured response drift, ideological consistency, and token-level sentiment shifts to develop comparative model scoring.

## Technical Skills

**Languages:** Python, JavaScript, TypeScript, SQL, C++, Rust, Go

**AI/ML:** PyTorch, TensorFlow, LangChain, MCP, RAG, LLMs, NLP, Computer Vision, Prompt Engineering

**Tools/Platforms:** React, Playwright, SQLite, Docker, Git, AWS (EC2, Lambda), APIs, SONiC